

IS 302-2-202 (1992)

भारतीय मानक

(Reaffirmed 2014)

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घरेलू और समान विद्युत साधित्रों की सुरक्षा

भाग 2 विवरणात्मक अपेक्षाएँ

अनुभाग 202 विद्युत स्टोव

Indian Standard

SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 202 Electric Stoves

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BUREAU OF INDIAN STANDARDS

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Price Group 3

Electrical Appliances Sectional Committee, ETD 32

FOREWORD

This Indian Standard was adopted by the Bureau of Indian Standards, after the draft finalized by the Electrical Appliances Sectional Committee had been approved by the Electrotechnical Division Council.

This standard covers the general and safety requirements for electric stoves to ensure personal safety against electric shock, safety against the effects of excessive temperature and fire, and reliable operation.

This standard covers electric stoves having open element while electric hot plates in which the element is enclosed or embedded in a cast body or within tubular sheath(s) are covered by IS 365 : 1983 Specification for electric hot plates.

This standard is to be read in conjunction with IS 302-1 (1979) 'Safety of household and similar electrical appliances: Part 1 General requirements'. For the sake of convenience, the clauses of this standard correspond to those of IS 302-1 (1979). Instead of reproducing full text of each clause, clauses of IS 302-1 (1979) which are applicable (which means than relevant provision of the clause apply) or not applicable and the sub-clauses or portions thereof which are not applicable are indicated as under:

- a) In case of a clause where it is applicable, the wording used is 'This clause of IS 302-1 (1979) is applicable/not applicable'.
- b) In case of sub-clause or part thereof 'Not applicable'.

Wherever a sub-clause of IS 302-1 (1979) is to be replaced by a new text, it has been indicated as under.

'Replacement — followed by the new text'

Any addition to the existing provisions of a sub-clause of IS 302-1 (1979) has been indicated as under:

'Addition — followed by the text of the additional matter' Clauses/tables which are additional to those of IS 302-1 are numbered starting from 101 and additional sub-clauses are numbered with the main clause number followed by 101, 102 etc, for example 7.101. Should however any deviation exist between IS 302-1 (1979) and this standard, the provisions of the latter shall apply.

Composite standards covering safety and performance requirements are covered by a separate standard IS : 2994-1992 'Specification for electric stoves'.

While preparing this specification, assistance has been derived from the following:

CCE 11-1954 Electric cooking and heating appliance for domestic and similar purposes. International Commission on Rules for the Approval of Electrical Equipment.

SABS : 153-1958 Electric stoves and hot plates. South African Bureau of Standards.

BS 3456 : Section A-1-1961 Heating and cooking appliances : Section A-1 General requirements. British Standards Institution.

IEC 335-1 (1976) Safety of household and similar electrical appliances — Part 1 General requirements, issued by International Electrotechnical Commission.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the results of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (revised)'. The number of significant places retained in the rounded off values should be the same as that of the specified value of this standard.

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SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 202 Electric Stoves

1 SCOPE

1.0 This clause of IS 302-1 (1979) is applicable except as follows:

1.1 This standard covers the general and safety requirements and methods of test for electrically-heated stoves containing open type heating elements, designed for connection to supplies at voltages not exceeding 250 V, ac single-phase 50 Hz or dc.

1.1.1 This standard does not cover multi-unit type electric stoves having more than one heating unit.

2 TERMINOLOGY

2.1 This clause of IS 302-1 (1979) is applicable except as follows:

2.8, 2.11, 2.18, 2.19, 2.21, 2.23, 2.25 to 2.28 not applicable.

2.29 Replacement

Conditions of Adequate Heat Discharge — The condition of adequate heat discharge denote that the electric stove is operated under the following condition:

A pan shall be positioned centrally over the heating surface and it shall have sufficient areas to cover the heating surface. The dimensions of the pan shall not exceed the dimensions of the heating surface by more than 10 mm. The pan shall be of normal, not brightly polished aluminium, the bottom surface being flat to within 0.05 mm.

The capacity of the pan shall be 3.0 litres per kilowatt rating and it shall contain 1.5 litres per kilowatt of initially cold water evaporation losses being made up as necessary. The test vessel should be covered by a well fitting lid.

2.30, 2.33, and 2.34 not applicable.

Additional Sub-clause

2.101 Electric Stove — An electric appliance containing open type heating element, designed to accommodate, on its upper surface, a vessels for heating liquids or cooking food.

3 GENERAL REQUIREMENTS

3.1 This clause of IS 302-1 (1979) is applicable except that the provisions of second para of **3.1** are not applicable.

4 GENERAL NOTES ON TESTS

This clause of IS 302-1(1979) is applicable except as follows.

4.2 Replacement

The type tests are carried out on the samples which shall withstand the relevant tests specified in the standard.

NOTES

1 If it is evident from the design of the electric stove that a particular test is not applicable, the test is not made.

2 Component such as switches required to be tested shall be submitted along with the electric stove.

4.8, 4.10 and 4.14 not applicable.

5 RATING

This clause of IS 302-1 (1979) is applicable except as follows.

5.2 Rated Input — The rated input of the heating element of an electric stove shall be one of the following:

750, 1 000 and 1 500 W.

NOTE — In case there are more than one heating element, even then, the total rating shall not exceed 1 500 W.

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6 CLASSIFICATION

This clause of IS 302-1 (1979) is applicable.

6.1 (a) (3), **6.1** (b) (2), (3) (4) not applicable.

7 MARKING

This clause of IS 302-1 (1979) is applicable except as follows:

7.1 Replacement

Each electric stove shall be marked with

- a) rated voltage/rated voltage range,
- b) symbol for nature of supply,
- c) rated frequency or rated frequency range in Hz,
- d) rated input in watts or kilowatts,
- e) manufacturer's name, trade-mark or identification mark,
- f) manufacturer's model or type reference,
- g) symbol for class II construction if of class 11 only,
- h) thermal efficiency, and
- j) country of origin.

NOTE — The rated input is total input of the circuits that can be in operation simultaneously.

7.6, 7.8 Not applicable.

Additional Sub-clause

7.101 The electric stoves may also be marked with the Standard Mark.

8 PROTECTION AGAINST ELECTRIC SHOCK

This clause of IS 302-1 (1979) is applicable except as follows:

8.2 to 8.4, and **8.7** not applicable.

9 STARTING OF MOTOR OPERATED APPLIANCES

This clause of IS 302-1 (1979) is not applicable.

10 INPUT AND CURRENT

This clause of IS 302-1 (1979) is applicable except as follows:

10.1 When electric stoves are fitted with or are intended to be used in conjunction with thermostats or thermal cut-outs, the input is measured immediately before the cut-outs

operate for the first time after an operating period of 30 minutes.

10.2 Not applicable.

11 TEMPERATURE RISE

This clause of IS 302-1 (1979) is applicable except as follows.

11.4 Replacement

Electric stoves are operated in accordance with the conditions of adequate heat discharge with all the heating elements that can be switched on at the same time being in circuit, the supply voltage being such that the input is 1.15 times the highest value of the sum of the rates input that can be switched on at the same time.

11.5 and 11.6 Not applicable.

11.7 Replacement

The electric stove is operated under conditions of adequate heat discharge until steady state conditions are established or for 60 minutes, whichever is earlier. The control devices are at the highest setting until the water boils and then the control devices are adjusted so as to maintain the water gently boiling, the pan being covered with the lid.

11.8 Addition

- i) The temperature of accessible exterior surface below a plane 50 mm below the top hob surface of the electric stove shall not exceed 150°C, and
- ii) Temperature-rise of the walls of the test corner shall not exceed 65°C.

11.9 and 11.10 Not applicable.

12 OPERATION UNDER OVERLOAD CONDITIONS OF APPLIANCES WITH HEATING ELEMENTS

This clause of IS 302-1 (1979) is applicable except as follows:

12.3 Not applicable.

13 ELECTRICAL INSULATION AND LEAKAGE CURRENT AT OPERATING TEMPERATURE

This clause of IS 3024 (1979) is applicable.

14 RADIO AND TELEVISION INTERFERENCE SUPPRESSION

This clause of IS 302-1 (1979) is not applicable.

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15 MOISTURE RESISTANCE

This clause of IS 302-1 (1979) is applicable except as follows:

15.1 to 15.3 Not applicable.

16 INSULATION RESISTANCE AND ELECTRIC STRENGTH (AFTER HUMIDITY TREATMENT)

This clause of IS 302-1 (1979) is applicable except as follows:

16.3 Not applicable.

17 OVERLOAD PROTECTION

This clause of IS 302-1 (1979) is not applicable.

18 ENDURANCE (Under Consideration)

19 ABNORMAL OPERATION

This clause of IS 302-1 (1979) is applicable except as follows:

19.1 Addition

If the heating element ruptures during the test, the test is repeated on a new sample which has been subjected to tests up to and including 12.

19.2 Replacement of the note under the clause by the following:

NOTE — The conditions without adequate heat discharge correspond to operation of the electric stove without pan. In case, there is control device, it shall be at the highest setting.

19.4 to 19.10 Not applicable.

20 STABILITY AND MECHANICAL HAZARDS

This clause of IS 302-1 (1979) is applicable except as follows:

20.2 Not applicable.

21 MECHANICAL STRENGTH

This clause of IS 302-1 (1979) is applicable except as follows:

21.1 Addition

Electric stoves employing glass or ceramic as essential for the enclosure of live parts shall withstand without cracking or breaking the impact of a utensil loaded with shot to a mass of 1.8 kg and dropped from a height of 150 mm. In the absence of other indications, a

vessel having a copper or aluminium bottom, flat over a diameter of 11 to 13 cm, with a corner radius of at least 8 cm shall be considered appropriate. The utensil is dropped flat ten times on to the representative heater areas of the surface.

21.2 to 21.4 Not applicable.

22 CONSTRUCTION

This clause of IS 302-1 (1979) is applicable except as follows:

22.5, 22.8, 22.9, 22.12, 22.15, 22.16, 22.22, 22.31 to 22.33 not applicable.

Additional Sub-clauses

22.101 The heating surface shall be in one plane.

22.102 The heating elements shall be so designed and supported that they maintain their intended position in the stove. It shall not be possible for a broken element to fall out of the stove or for an element to become displaced when the stove is lifted by the carrying handle.

22.103 Heating elements of replaceable type shall be so designed and constructed as to be easily replaceable.

22.104 When selecting materials for connecting heating element terminations, care shall be taken to choose those which are compatible with the type of resistance material used. Terminals and terminal parts of the heating elements shall be nickel-plated or otherwise protected from corrosion unless made from non-corroding material.

22.105 Heating elements shall be protected by means of metal guards to prevent any inadvertent contact by person, or with any object adjacent to the electric stove. The purpose of the guards is to reduce or prevent the risk of fire or electric shock due to accidental contact with or proximity to heating element.

22.105.1 The guards shall be of robust construction and securely attached to the electric stove. If the guard is so designed and attached that the whole or any portion of the guard can be removed, lifted or swung to one side for maintenance purposes, the methods of attachment shall be such that the guard, when replaced, cannot be accidentally dislodged from the working position during normal use.

22.105.2 It shall not be possible to touch the wires or terminations of the heating element or

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their contact members with any part of a test finger conforming to IS 1401: 1970 'Accessibility test probes (*first revision*)', when such a probe is inserted through or around the guard without undue pressure.

22.106 Electric stoves shall be provided with a handle or other suitable means of handling, securely attached to it. Grip of the handle provided, shall be of a material electrically non-conducting and heat insulating.

22.107 Switches and wiring shall be so situated or protected that they are unaffected by steam, vapour or spillage produced during normal cooking operation.

22.107.1 Control switches, if provided, shall be so located that, in normal uses they are easily accessible, but are not exposed to damage.

23 INTERNAL WIRING

This clause of IS 302-1 (1979) is applicable except as follows:

23.4 to 23.7 Not applicable.

24 COMPONENT

This clause of IS 302-1 (1979) is applicable except as follows:

24.4 to 24.10 Not applicable.

Additional Sub-clause

24.101 Refractory material if used shall be of adequate mechanical strength and durability as to withstand the mechanical shock during other test specified in the standard. It shall not deteriorate by alternate heating and cooling normal in service.

25 SUPPLY CONNECTIONS AND EXTERNAL FLEXIBLE CABLES AND CORDS

This clause of IS 302-1 (1979) is applicable except as follows:

25.1 Replacement

Electric stoves shall be provided with a power supply cord or an appliance inlet for connection to the supply. Compliance is checked by inspections.

25.2 Not applicable.

25.4 Addition

The length of the power supply cord external to the appliance shall be not less than 2 mm.

25.7 Not applicable.

25.10 Addition

Provision of cord guard is optional for electric stoves.

25.13 Not applicable.

26 TERMINALS FOR EXTERNAL CONDUCTORS

This clause of IS 302-1 (1979) is applicable.

27 PROVISIONS FOR EARTHING

This clause of IS 302-1 (1979) is applicable.

28 SCREWS AND CONNECTIONS

This clause of IS 302-1 (1979) is applicable.

29 CREEPAGE DISTANCES AND CLEARANCES

This clause of IS 302-1 (1979) is applicable.

30 RESISTANCE TO HEAT FIRE AND TRACKING

This clause of IS 302-1 (1979) is applicable.

31 RESISTANCE TO RUSTING

This clause of IS 302-1 (1979) is applicable except as follows:

31.1 Replacement

Ferrous parts, rusting of which might cause the appliance to fail to comply with this standard, shall be adequately protected against rusting.

32 RADIATION HAZARDS

This clause of IS 302-1 (1979) is not applicable.

101 TESTS

101.0 Categories of Tests — Tests are classified as type, routine and acceptance tests.

101.1 Type Tests — The tests specified in Table 101 shall constitute the type tests and shall be carried out on two samples of electric stoves of the same type and rating selected preferably at random from a regular production lot. Before commencement of the tests, the electric stoves shall be visually examined and inspected for obvious visual defects in respect of components, parts and their assembly, construction, stability, markings, provision of suitable terminals for supply connections, earthing and the effectiveness of screws and connections. The external surface finish shall be even and free from finishing defects.

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Table 101 Schedule of Type Tests
(Clause 101.1)

Sl No.	Test	Clause Reference
1.	Protection against electric shock	8
2.	Input	10
3.	Temperature rise	11
4.	Operation under overload conditions	12
5.	Electrical insulation and leakage current at operating temperature	13
6.	Moisture resistance	15
7.	Insulation resistance and electric strength (after humidity treatment)	16
8.	Endurance (<i>under consideration</i>)	18
9.	Abnormal operation	19
10.	Stability and mechanical hazards	20
11.	Mechanical strength	21
12.	Construction	22
13.	Internal wiring	23
14.	Components	24
15.	Cord grip and cord guard	25
16.	Terminals for external conductors	26
17.	Provision for earthing	27
18.	Screws and connections	28
19.	Creepage distance and clearances	29
20.	Resistance to heat, fire and tracking	30
21.	Resistance to rusting	31

101.1.1 Criteria of Acceptance — Both samples shall successfully pass all the type tests for proving conformity with the requirements of the standard. If any of the samples fails in any of the type tests, the testing authority at its discretion, may call for fresh samples not exceeding twice the original number and subject

them again to all tests or to the test(s) in which failure(s) occurred. No failure shall be permitted in the repeat test(s).

101.2 Acceptance Tests — The following shall constitute the acceptance tests:

Test	Clause Reference
a) Protection against electric shock	8
b) Input	10
c) Temperature rise	11
d) Insulation resistance and electric strength at operating temperature	13
e) Moisture resistance	15
f) Insulation resistance and electric strength (after humidity treatment)	16
g) Earthing connection	27

101.2.1 A recommended sampling procedure for acceptance tests is given in Appendix B of IS 302-1 (1979).

101.3 Routine Test — The following test shall constitute the routine tests:

Test	Clause Reference
a) Protection against electric shock	8
b) High voltage	13.3.2 of IS 302-1 (1979)
c) Earthing connection	27

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ANNEX A

TABLES OF TYPE TESTS

Appendix A of IS 302-1 (1979) is not applicable.

ANNEX B

SAMPLING PROCEDURE FOR ACCEPTANCE TESTS

Appendix B of IS 302-1 (1979) is applicable.

ANNEX C

ELECTRONIC CIRCUITS

Appendix C of IS 302-1 (1979) is applicable if electronic circuits are used.

ANNEX D

MEASUREMENT OF TEMPERATURE WITH THERMOMETER

Appendix D of IS 302-1 (1979) is applicable.

ANNEX E

ALTERNATIVE TESTS FOR PROTECTED MOTOR UNITS

Appendix E of IS 302-1 (1979) is not applicable.

ANNEX F

IMPACT TEST APPARATUS

Appendix F of IS 302-1 (1979) is applicable.

ANNEX G

THERMAL CONTROLS AND OVERLOAD RELEASES

Appendix G of IS 302-1 (1979) is applicable if thermal controls and overload releases are used.

ANNEX H

MEASUREMENT OF CREEPAGE DISTANCES AND CLEARANCES

Appendix H of IS 302-1 (1979) is applicable.

ANNEX J

TEST FOR FIRE-RESISTING PROPERTIES

Appendix J of IS 302-1 (1979) is applicable.

ANNEX K

**BNF JET TEST FOR DETERMINATION OF THICKNESS OF COPPER AND NICKEL
PLATING**

Appendix K of IS 302-1 (1979) is not applicable.

ANNEX M

**APPROXIMATE MEASUREMENT OF THICKNESS OF CHROMIUM ON NICKEL,
STEEL AND COPPER**

Appendix M of IS 302-1 (1979) is not applicable.

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